

Impact of Maximal Overexpression of a Non-toxic Protein on Yeast Cell Physiology

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Abstract

Summary

While it is recognized that excess expression of non-essential proteins burdens cell growth, the physiological state of cells under such stress is largely unknown. This is because it is challenging to distinguish between adverse effects arising from the properties of the expressed excess protein (cytotoxicity) and those caused solely by protein overexpression. In this study, we attempted to identify the model protein with the lowest cytotoxicity in yeast cells by introducing a new neutrality index. We found that a non-fluorescent fluorescent protein (mox-YG) and an inactive glycolytic enzyme (Gpm1-CCmut) showed the lowest cytotoxicity. These proteins can be expressed at levels exceeding 40% of total protein while maintaining yeast growth. The transcriptome of cells expressing mox-YG to the limit indicated that the cells were in a nitrogen source requirement state. Proteome analysis revealed increased mitochondrial function and decreased ribosome abundance, like the inactivated state of the TORC1 pathway. The decrease in ribosome abundance was presumably due to defective nucleolus formation, partially rescued by a mutation in the nuclear exosome. These findings suggest that massive overexpression of excess protein, termed protein burden, causes nitrogen source starvation, a metabolic shift toward more energy-efficient respiration, and a ribosomal biosynthesis defect due to an imbalance between ribosomal protein and rRNA synthesis in the nucleolus.

eLife assessment

This **convincing** study advances our understanding of the physiological consequences of the strong overexpression of non-toxic proteins in baker's yeast. The findings suggest that a massive protein burden results in nitrogen starvation and a shift in metabolism likely regulated via the TORC1 pathway, as well as defects in ribosome biogenesis in the nucleolus. The study presents findings and tools that are **important** for the cell biology and protein homeostasis fields.

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Introduction

Expression levels of proteins are optimized to maximize the fitness of organisms, and the evolutionary principle that determines those levels is considered to be demand and constraint (Bruggeman et al., 2020) (Figure 1A). Demand is the evolutionary pressure that works to increase the expression of a protein to meet the requirements of its function, and constraint is the evolutionary pressure that works to decrease the expression of a protein to avoid the negative effects produced by excess protein. Constraints appear as disturbances in cellular function when the protein is overexpressed (Figure 1A, the red arrow). The mechanisms of constraining expression levels, or defects due to overexpression, can be roughly classified into four major categories (Moriya, 2015) (Figure 1B). “Resource overload” (also called “burden”) (Kastberg et al., 2022) is a disorder in which intracellular resources used for protein processing, such as synthesis, localization, degradation, and modification of proteins, are depleted or monopolized by the processing of that protein, resulting in a disruption of processing of other proteins (Dong et al., 1995; Eguchi et al., 2018; Kintaka et al., 2020, 2016; (Stoebel et al., 2008). This phenomenon occurs in proteins with high expression levels. “Stoichiometry imbalance” is a disorder triggered by an imbalance of amounts between constituents within complexes, leading to untimely activation or inactivation (pathway modulation), degradation (resource overload), aggregation (promiscuous interactions), etc. (Kaizu et al., 2010; Makanae et al., 2013; Papp et al., 2003). This occurs due to the overexpression of proteins that form complexes. “Pathway modulation” is a disorder in which pathways are untimely activated or inactivated. This occurs especially with overexpression of proteins involved in pathway regulation, such as metabolic enzymes, kinases and phosphatases, and transcription factors (Prelich, 2012; Youn et al., 2017). “Promiscuous interaction” is a disorder in which weak interactions are enhanced by mass action to form complexes or aggregates that are not normally formed. This damages cellular function by inactivating essential proteins or overloading degradation resources, and is caused by overexpression of intrinsically disordered proteins or liquid-liquid phase-separating proteins (Bolognesi et al., 2016; Ma et al., 2010; Vavouri et al., 2009).

As the expression level of a protein increases, it eventually encounters one of the barriers created by either these specified constraints or others that are unknown, all of which are related to the function and physicochemical properties of the protein, and damage cellular function. The expression level at which this occurs is referred to as the “expression limit” in this paper (see Figure 1A). If the constraints could be removed from the protein, the expression limit would increase. For example, if components of a complex are simultaneously expressed to resolve a stoichiometry imbalance, the expression limits of those proteins will be increased (Kaizu et al., 2010; Makanae et al., 2013). In this manner, when (one) constraint is removed from a protein, the expression limit would increase until the next constraint barrier is hit (Figure 1A, dotted lines). The amount of expression limit created by a constraint barrier depends on the type of constraint. Stoichiometry imbalances, pathway modulations, and promiscuous interactions would create different expression limits for each protein. This is because they are determined by the activity, regulatory mechanism, and physicochemical properties of each protein. On the other hand, resource overload is presumed to be determined to some extent by the capacities of the processes in the cell (Kintaka et al., 2020) (Figure 1C). Essentially, the synthesis, in which all proteins are processed, should have the largest capacity, while processes such as transport, folding, and degradation, in which only a fraction of proteins are processed, should have smaller capacities. In fact, adding transport or degradation signals to a cytosolic protein lowers the expression limit (Kintaka et al., 2016). In that light, the final constraint barrier that emerges when all possible constraints are removed is the overloading of the synthesis process.

Overloading of the synthesis process is specifically referred to as “protein burden” or “protein cost” (Eguchi et al., 2018 [↗](#); Kafri et al., 2016 [↗](#)). It has long been the subject of study, mainly as an effect on growth produced by excess (gratuitous) proteins that are not necessary for cellular function (Bruggeman et al., 2020 [↗](#); Scott et al., 2010 [↗](#)). This idea was first proposed in studies of the cost of unnecessary induction of the *lac* operon in *Escherichia coli* (Koch, 1983 [↗](#)). In recent years, much attention has been paid to the actual mechanisms by which massive expression of excess protein inhibits growth and the response of cells (Farkas et al., 2018 [↗](#); Kafri et al., 2016 [↗](#); Metzl-Raz et al., 2020 [↗](#)). During protein burden, it is presumed that depletion of the materials, machinery, and factors necessary for transcription and translation occurs, along with competition from these monopolies. In particular, amino acids and aminoacyl-tRNAs are major candidates for depletion (Kastberg et al., 2022 [↗](#)). In addition, ribosomes, which are crucial for protein synthesis, themselves require significant resources for their synthesis and have therefore been considered major candidates for depletion during protein burden, as they compete for these limited resources (Metzl-Raz et al., 2017 [↗](#)). Mechanistic analysis of protein burden in budding yeast, a model eukaryotic cell, has been intensively conducted by Barkai and colleagues, who reported that transcription is rate-limiting in phosphate starvation and translation in nitrogen source starvation (Kafri et al., 2016 [↗](#)), and that basic transcription factors are limiting resources (Metzl-Raz et al., 2020 [↗](#)). In addition, a cellular response in the form of increased cell size and enhanced protein synthesis capacity has been observed (Kafri et al., 2016 [↗](#)). Thus, the previously expected decrease in ribosomes has not been detected.

As mentioned above, to investigate protein burden, we need to overexpress proteins with minimal constraints. This is because overexpressing proteins with additional constraints cannot overload the synthetic process, as the additional constraints reduce the expression limits (**Figure 1A** [↗](#)). Also, the cellular damage and cellular response generated by the constraint mask the protein burden response. However, it remains unclear which proteins are not subject to constraints other than protein burden. Proteins that would induce protein burden have been determined by assuming that they would be minimally constrained based on their properties, and this assumption has been confirmed experimentally. Model proteins selected in this manner in yeast include fluorescent proteins (FPs) and certain glycolytic enzymes (Eguchi et al., 2018 [↗](#); Farkas et al., 2018 [↗](#); Kafri et al., 2016 [↗](#); Kintaka et al., 2020 [↗](#), 2016 [↗](#)). FPs do not have a specific function or interaction partner in yeast cells and are not localized. On the other hand, glycolytic enzymes are naturally expressed at very high levels and are thus considered to be less constrained. Notably, these proteins exhibit an expression limit of approximately 15% of total protein in yeast cells (Eguchi et al., 2018 [↗](#); Kintaka et al., 2016 [↗](#)). However, it is not clear whether these proteins are free from constraints other than protein burden. This is because there may be unknown constraints, and there is no evidence that the measured expression limit is the “true limit” at which protein burden occurs.

In addition to the protein (amino acid sequence) itself, there is another aspect to be considered when selecting a model for protein burden. That is the nucleotide sequence of the gene used for expression, i.e., transcriptional, and translational regulatory sequences, codon usage, and so on (**Figure 1D** [↗](#)). For example, if a protein is to be expressed to the limit by a less codon-optimized and thus less translationally efficient mRNA, a large amount of mRNA would be required, which would induce a greater burden on transcription. Conversely, a similar attempt by an mRNA with high translation efficiency would induce a greater burden on translation. These are the constraints inherent in protein synthesis itself. Although there is some evidence that protein synthesis in nature is economically designed to optimally load transcription and translation (Frumkin et al., 2017 [↗](#); Hausser et al., 2019 [↗](#); Mahima and Sharma, 2023 [↗](#)), it still needs to be determined what the genetic design should be to minimize the burden.

Considering the above circumstance, a realistic approach to protein burden requires more than just drawing straightforward conclusions from the overexpression of a single model protein (or gene). Rather, it requires experimentally identifying proteins with the highest possible expression

limits, investigating the negative impacts of their overexpression, and, if any constraints are revealed, examining whether their removal can increase expression limits. Through successive iterations, this process gradually deepens our understanding of protein burden (**Figure 1E**). In each iteration, the least constrained protein (and its gene) will be generated and utilized as a benchmark for further investigation into proteins with high (or low) expression limits and the adverse effects of their overexpression. This will (unintentionally) elucidate a new mechanism of constraint. Indeed, the enhanced green fluorescent protein (EGFP), which we have used as a model for protein burden (Kintaka et al., 2020, 2016; Makanae et al., 2013), was found to form a chaperone-entrapped aggregate upon overexpression and trigger the heat shock response due to its low folding ability (Namba et al., 2022). Its overexpression also causes abnormal yeast cell morphology through proteasomal stress as it contains cysteines. The moxGFP (mox), which improved folding properties and lacks cysteine residues (Costantini et al., 2015), showed a higher expression limit and no longer induced abnormal morphologies. However, even when overexpressing mox, the anticipated depletion and competition for factors, or cellular responses to them have not been observed (Namba et al., 2022).

In this study, we found that a mutation causing loss of fluorescence in mox, and other fluorescent proteins increased the expression limit up to 3-fold, to more than 40% of total protein. Introducing a neutrality index (NI) to assess protein constraint, we found that non-fluorescent mox (mox-YG) and enzymatically inactive Gpm1 (Gpm1-CCmut) are currently the least constrained. Upon overexpression of mox-YG, depletion of amino acid or nitrogen sources, decreased ribosomal expression, and a metabolic shift from glycolysis to oxygen respiration were observed. Thus, physiological states, as anticipated with such protein burdens, are only generated by massive overexpression of proteins with significantly lower constraints. Some of the responses found appear to be due to inactivation of the TORC1 pathway, suggesting that eukaryotic cells use this regulatory pathway to buffer protein burden. In the burdened cells, the nucleolus was dysplastic, and mutations in the nuclear exosome partially restored this. Thus, the abnormal formation of liquid-liquid phase-separated organelles due to dilution may be one of the defects caused by the strong protein burden.

Results

Fluorescent property is a constraint on the expression of fluorescent proteins

To evaluate the expression limits of FPs, we overexpressed FPs to levels that induced growth inhibition. Gene constructs, in which the *TDH3* promoter controls the target protein genes, were incorporated into plasmids designed for the genetic-tug-of-war (gTOW) method (**Figures 2A**, **2_S1A**) (Moriya et al., 2006). Subsequently, yeast cells were transformed and cultured in a medium depleted of leucine. The maximum growth rate (MGR) of cells was calculated from serial measurements of optical density at 660 nm (OD_{660}). Protein levels were determined through the analysis of proteins from cells in the logarithmic growth phase (**Figure 2_S1B**). For each protein, we also analyzed a mutant in which the Tyr residue forming the fluorophore was replaced with Gly (YG mutation), resulting in a loss of fluorescence. This is because we unexpectedly found that this mutation substantially increased the expression limits of FPs (see below).

Overexpression of the FPs and mutants all caused growth inhibition (**Figure 2B**, **C**). EGFP and mox were expressed at about 15% of the total protein as previously reported (Kintaka et al., 2016), but mCherry (mChe) and YG mutants were expressed at higher levels. Mox-YG expressed 44% of the total protein (**Figures 2D**, **E**). Our previous work had evaluated only expression limits (Eguchi et al., 2018), but we realized that we could not properly assess the constraints on each protein if the growth rates at the expression limit were different. For example, the mChe's expression limit (20%) is higher than that of sfGFP (17%), but the relative MGRs are 20% and 80%,

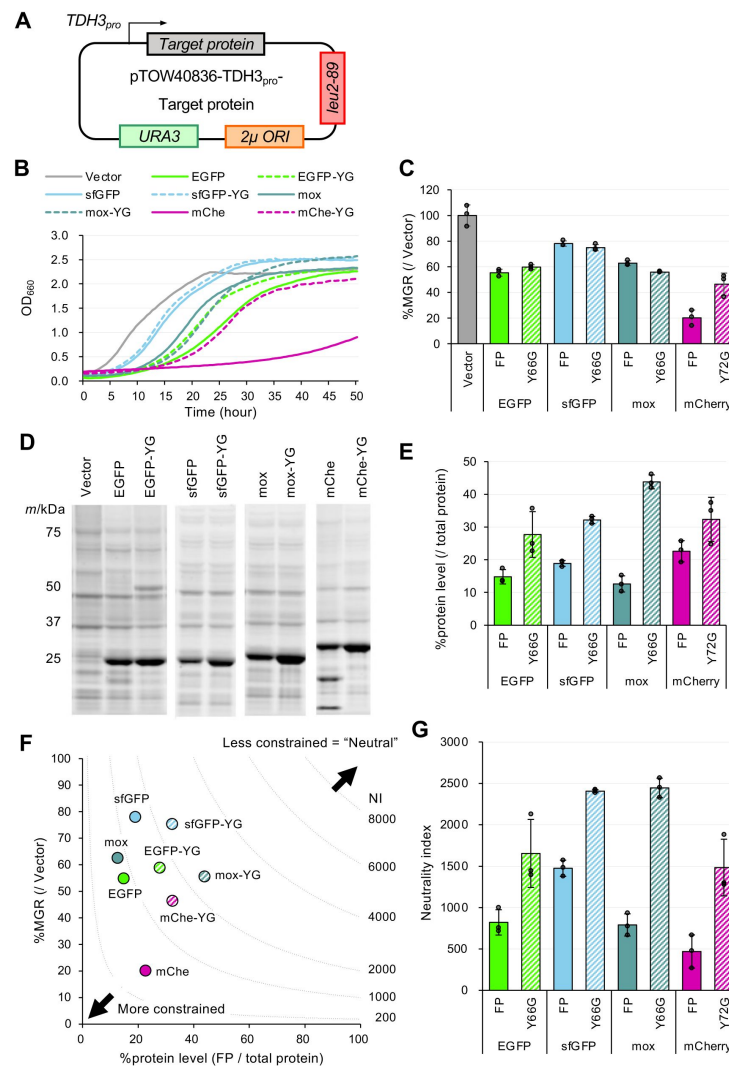


Figure 2.

Evaluation of expression constraint (or neutrality) of fluorescent proteins.

A) Experimental setup of the analysis. Target proteins were expressed under the control of the *TDH3* promoter (*TDH3_{pro}*) using the multicopy plasmid pTOW40836. The cells were pre-cultured under –Ura conditions, and then cultured in –LeuUra conditions.

B) Growth curves of the vector control and cells overexpressing FPs and their mutants in synthetic medium (–LeuUra). The solid or dotted lines show the average calculated from three biological replicates. Growth curves with the standard deviations (SD) of replicates are shown in [Figure 2 S2](#).

C) MGR of cells overexpressing FPs and their mutants (percent over the vector control).

D) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing FPs. All proteins were separated after staining with a fluorescent dye.

E) Protein level of the target protein (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of D. See [Figure 2_S1B](#) for the method of FP and total protein quantification. Note that EGFP-YG and mCherry may not be accurately measured due to dimerization and cleavage of the protein, respectively.

F) Relationship between MGR and the protein level. MGR data are the same as in **C**, and the protein level data are the same as in **E**. The neutrality indexes (NIs), calculated from the products of MGR and protein levels, are indicated by the lines. If the investigated proteins have the same NI, they are located on the same dotted line segment.

G) The neutrality index for the FPs and mutants. The neutrality index is the product of %MGR (/vector) and %protein level (/total protein).

The bars and error bars in **C**, **E**, and **G** show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots.

respectively, which is much lower for mChe. The relationship between %MGR and %protein level when the vector is set at 100% is shown in the scatter plot in **Figure 2F**. The lower left indicates more constraint as lower expression levels lead to growth inhibition, and the upper right indicates less constraint as higher expression levels do not cause growth inhibition. Here, we describe “unconstrained” as “neutral” and calculate the neutrality index (NI) as the product of %MGR and %protein level.

Note that the regime of NI in **Figure 2F** shows a theoretical neutrality distribution but does not show the relationship between growth rate and the expression level when the expression level of a protein with a particular neutrality is increased or decreased. For example, 20% of the growth rate would be maintained when the expression of mox-YG (NI = 2,400) is increased to 100%, but this is not realistic.

The NI of each protein indicates that introducing a non-fluorescent mutation in FPs uniformly increases neutrality (**Figure 2G**), suggesting that “having fluorescence” is a constraint on the expression of FPs. The most neutral proteins were sfGFP-YG and mox-YG, with NIs of about 2,400. Notably, despite significant differences in the degree of growth inhibition and expression limits between the two proteins, their NIs are nearly identical (**Figures 2F, G**). They differ only in two amino acid residues (Cys - Ser) (**Figure 2, S3**), which seem to generate differences in expression efficiency. However, this does not appear to affect their NIs.

We also investigated the impact of expression efficiency on NI with another FP, mChe. The mChe used in this study is codon optimized (the codon adaptation index (CAI) = 0.584) (Keppler-Ross et al., 2008), while the mChe used by Kafri et al. for protein cost analysis (here referred to as mChe-Kafri) is not codon optimized (CAI = 0.236) (Kafri et al., 2016). When overexpressed, mChe-Kafri (and mChe-Kafri-YG) had lower expression and growth inhibition than mChe (and mChe-YG) (**Figures 2, S4A-D**). This may be due to the lower translation rate of mChe-Kafri mRNA. However, their NIs were not significantly different, especially for the YG mutation (**Figures 2, S4E, F**). Thus, the constraints indicated by NI are apparently not affected by translation efficiency. In the study using this mCherry-Kafri, it was also reported that the more mChe-Kafri was expressed, the more the cells enlarged (Kafri et al., 2016). However, no cell enlargement was observed with mox-YG, which can be expressed more than mCherry. Interestingly, the mChe-YG cells did not show enlargement (**Figures 2, S5A, B**), indicating that cell enlargement is a phenomenon that occurs specifically when fluorescent mChe is overexpressed.

It should be noted that the growth inhibition caused by the overexpression of mox-YG in this study is a burden on translation rather than on transcription. This is because overexpression of a frameshift mutant, in which four nucleotides were inserted directly after the ATG of the mox-YG gene, abolished the notable growth delay (**Figures 2, S5C, D**).

Constraint on fluorescent protein expression arises from fluorophore properties

Next, we further analyzed the possibility that fluorescence constrains the expression of FPs. When yeast was grown in the dark, the degree of growth inhibition of mox was the same (**Figure 2, S6A**) and the expression limit of mox was rather increased (**Figure 2, S6B**). Phototoxicity does indeed exist when expressing mox with fluorescence. This is because, when exposed to strong light, cells overexpressing mox show a reduced growth rate (**Figure 2, S6C-E**) and abnormalities in their internal structures (**Figure 2, S6F**). However, the expression limits of mox and mox-YG were almost unchanged by the light (**Figure S2, S6G, H**). Thus, the constraint was not due to phototoxicity itself.

The number of Tyr residues was also irrelevant because NI was like that of mox-YG when a single Tyr was added to the C-terminus of mox-YG (**Figure 2** [S7C-E](#)). Reactive oxygen species (H_2O_2) are generated during fluorophore formation. When H_2O_2 was added to the medium, stronger growth inhibition occurred only in mox (**Figure 2** [S7F-H](#)). This suggests that the reactive oxygen species generated during fluorophore formation are responsible for the constraint. However, this possibility is not supported by the omics analysis described later (see Discussion). Mox-YG, like mox, was present throughout the cytoplasm and did not cause any characteristic localization or aggregation (**Figure 2** [S8](#)).

During this study, we examined a moxGFP (mox-T203I) in which Thr203 was accidentally replaced by Ile (**Figure 2** [S9](#)). The fluorescence of this mutant is reduced to 78% of mox (**Figures 2** [S9A-C](#)). The growth rate of the cells expressing this mox-T203I was like that of mox and mox-YG (**Figure 2** [S9E](#)). Still, the expression limit was intermediate between mox and mox-YG (**Figure 2** [S9F](#)), resulting in the NI intermediate between mox and mox-YG (**Figure 2** [S9G](#)). This result further supports that fluorescence is related to constraints. However, as mentioned above, fluorescence emission itself should not be toxic (**Figure 2** [S6](#)). We next focused on the formation process of the fluorophore: the tyrosine of the GFP fluorophore undergoes phenol and phenolate state changes, with Thr203 stabilizing the phenol form and substitution to Ile reducing the phenolate form (Kummer et al., 2000 [S9](#)). Furthermore, in enhanced GFPs such as mox, substitution of Ser65 before Tyr66 for Thr anchors the phenolate form. To investigate the possibility that this is a factor determining the limit of mox, we produced mox with Thr65 replaced with Ser and examined its expression limit. The mox-T65S mutant was expressed 12% more than mox (**Figures 2** [S9H, I](#)), suggesting that the phenol-type fluorophore can be a limiting factor for the expression level of the fluorescent protein, while the concrete mechanism is still unclear.

The activities of the glycolytic enzymes Gpm1 and Tdh3 may constrain their expression

We previously investigated the expression limits of a group of glycolytic enzymes in *S. cerevisiae*, with Gpm1 and its catalytic center (CC-) mutant having the highest expression limits (Eguchi et al., 2018 [S9](#)). Here, we evaluated the neutrality of the wild types and their CC-mutants of Tdh3 and Gpm1 (**Figure 3** [S9](#)). The growth rates at the expression limits were higher in the CC-mutant for both Tdh3 and Gpm1, and in Gpm1, the CC-mutant grew two-fold faster (**Figures 3A** [S9](#), **B**, **3** [S1](#)). The protein levels of wild types and mutants of both proteins were not significantly different (**Figures 3C** [S9](#), **D**). The calculated NIs were highest for Gpm1 CC-mutant, which was equivalent to that of mox-YG (**Figures 3E** [S9](#), **F**). Thus, mox-YG and Gpm1 CC-mutant are the least constrained proteins to this point. We previously concluded that the activities of glycolytic enzymes generally do not affect their expression limits (Eguchi 2018), but their neutralities suggest that their activities can become constraints in some way. This may be due to a disturbance in metabolism or a property of the active enzyme, such as an effect of reactive amino acids (His or Cys).

Overexpression of mox-YG causes an amino acid starvation response

To understand the cell physiology under protein burden, we next performed transcriptome (RNA-seq) analysis of the cells overexpressing mox-YG, the least constrained protein. Although the expression limit of mox-YG is about 3-fold higher than that of mox (**Figure 2E** [S9](#)), the growth rates at the expression limits of both are not largely different (**Figure 2C** [S9](#)). Thus, by comparing transcriptomes upon their overexpression, we may be able to distinguish and extract the general transcriptional response due to reduced growth rates from the (protein burden) response associated with the massive expression of a less constrained protein.

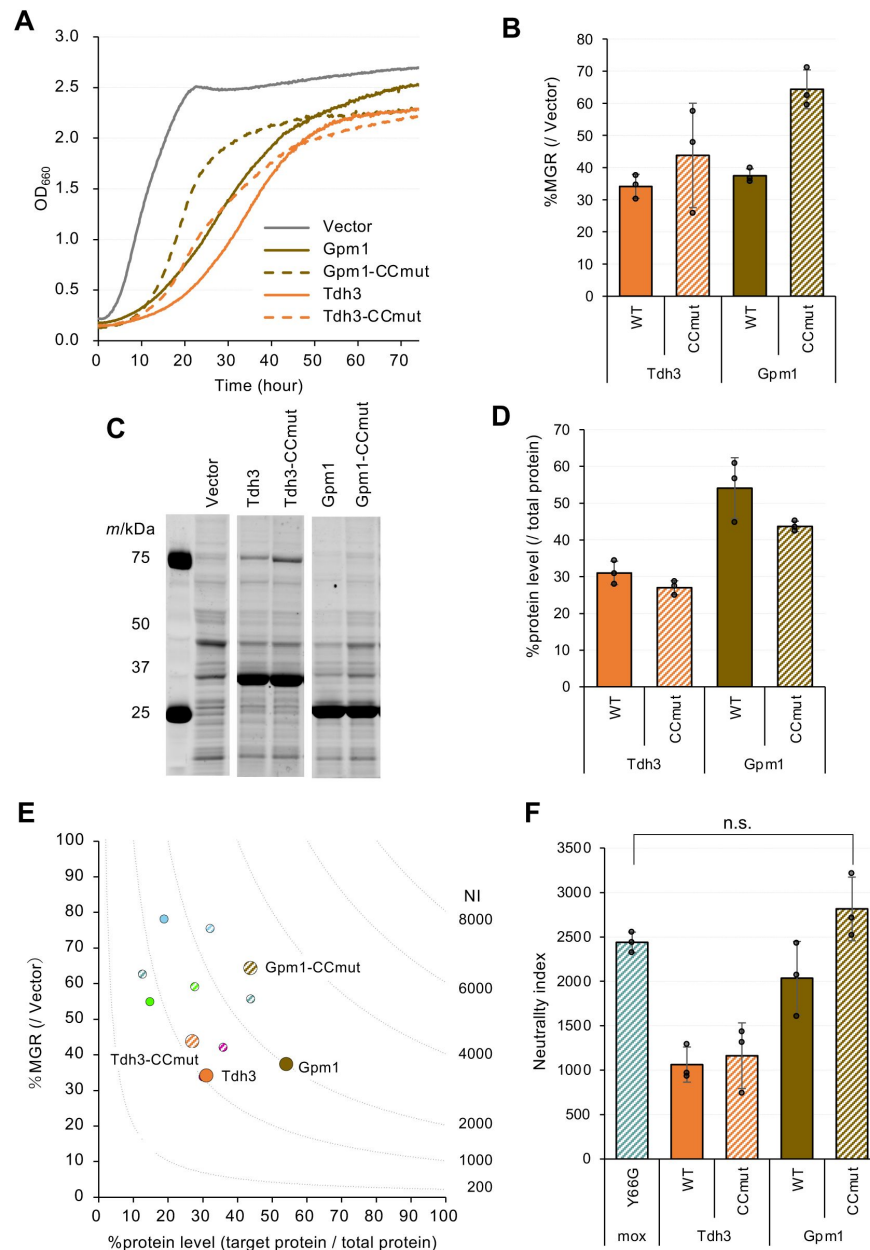


Figure 3.

Evaluation of the neutrality of Tdh3 and Gpm1.

A) Growth curves of cells with the control vector (Vector) and overexpressing Gpm1 and Tdp3 and their catalytic center mutants (CCmut) in synthetic medium (–LeuUra). The solid or dotted lines show the average calculated from three biological replicates. Growth curves with SDs of replicates are shown in **Figure S3-S1**.

B) MGR of cells overexpressing indicated proteins (percent over the vector control).

C) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing indicated proteins.

All proteins were separated after staining with a fluorescent dye.

D) Protein levels of the target protein (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of C.

E) Relationship between MGR and the protein level. MGR data are the same as in **B**, and the protein level data are the same as in **D**. The neutrality indices (NIs) are indicated by the lines in **Figure 2F**.

F) The neutrality index for Gpm1, Tdh3, and their CC-mutants.

The bars and error bars in B, D, and F show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots.

The expression changes under *mox* and *mox-YG* overexpression were calculated as fold change (FC) over the vector control (**Figures 4** [S1A, B](#)). The overall trend of expression changes was similar ($r = 0.64$, **Figure 4A** [S1](#)). This similar variation may include responses associated with the expression system and reduced growth rate. On the other hand, there were also different transcriptional responses between the two. Gene categories that showed significant expression changes in either *mox* or *mox-YG* overexpression are shown in **Figure 4B** [S1](#). In addition to the enhanced ribosome synthesis and decreased glycolysis observed in previous *mox* overexpression (Namba et al., 2022 [S1](#)), enhanced oxidative phosphorylation was detected. Interestingly, “Ribosome biogenesis”, which was elevated in *mox* overexpression, showed a rather decreasing trend in *mox-YG* overexpression (**Figure 4B** [S1](#)). Indeed, *IFH1*, which is involved in ribosomal gene transcription (Martin et al., 2004 [S1](#)), was upregulated in *mox*, but downregulated in *mox-YG* (**Figure 4A** [S1](#)). Different trends were also observed for amino acid metabolism (**Figure 4B** [S1](#)). The amino acid transporter genes *AGP1* and *GAP1* were upregulated only in *mox-YG* overexpression (**Figure 4A** [S1](#)). Expressions of specific amino acid and nitrogen source transporter genes were strongly induced in *mox-YG* (**Figure 4C** [S1](#)). The expression of glucose transporter genes was also very different (**Figure 4C** [S1](#)). The transcriptional responses in *mox-YG* overexpression obtained by RNA-seq were confirmed by transcriptional reporter analysis (**Figure 4D** [S1](#), **4_S2**). These results suggest that *mox* and *mox-YG* induce similar but distinct transcriptional responses, particularly that *mox-YG* overexpression may result in depletion of amino acid and nitrogen sources and decreased ribosome synthesis.

Overexpression of *mox-YG* causes a proteomic response that partially overlaps with TORC1 inactivation

Protein burden is an overload on synthetic resources, and thus, when this overload occurs strongly, it is expected that there would be a reduction in the amount of proteins other than those being overexpressed due to competition for synthetic resources. Indeed, in cells overexpressing *mox-YG*, the amounts of proteins other than *mox-YG* were reduced (**Figure 5A** [S1](#)). Interestingly, the total amount of protein per cell remained almost constant, with the amount of other proteins decreasing as *mox-YG* was overexpressed. This trend was generally true for the overexpression of other FPs and glycolytic enzymes (**Figure 5** [S1](#)). This suggests that there is a mechanism (or constraint) that maintains the overall cellular protein levels. To examine changes in protein expression, proteomic analysis of *mox-YG* overexpressing cells was performed. The distribution of intensities obtained from the analysis is shown in the Proteomaps in **Figure 5B** [S1](#). In *mox-YG* overexpressing cells, *mox-YG* accounted for 40% of the total protein expression. **Figure 5A** [S1](#) suggests that in *mox-YG* overexpressing cells, proteins other than *mox-YG* are overall reduced to about 60% compared to the vector. Therefore, we considered it necessary to normalize the data overall to accurately represent changes in the per-cell proteome. However, in this study, we used data without such correction to focus on how the composition of the proteome, other than *mox-YG*, changed (see Materials and Methods and **Figures 5** [S1, S2](#) for more details).

Figure 5C [S1](#) illustrates the categories that showed significant changes upon *mox-YG* overexpression (all categories are shown in **Figure 5** [S3](#)). Elevated mitochondrial functions, such as oxidative phosphorylation and TCA cycle, and decreased ribosomal protein and ribosomal synthesis were observed. As mentioned above, RNA-seq results suggested nitrogen source starvation in *mox-YG* overexpression. Therefore, we investigated whether this response is influenced by the regulation through the TORC1 pathway, a primary mechanism responsive to nitrogen sources. By comparing expression changes due to rapamycin (a TORC1 inhibitor) treatment (Gowans et al., 2018 [S1](#)) and *mox-YG* overexpression, we found that about 40% of the proteins with altered expression were congruent with the group of proteins regulated by TORC1 (**Figures 5D** [S1](#), **E**). Thus, the proteome expression changes due to *mox-YG* overexpression may include responses due to inactivation of the TORC1 pathway as well as other responses. **Figure 5C** [S1](#) shows which categories had significant expression changes when separated into rapamycin-

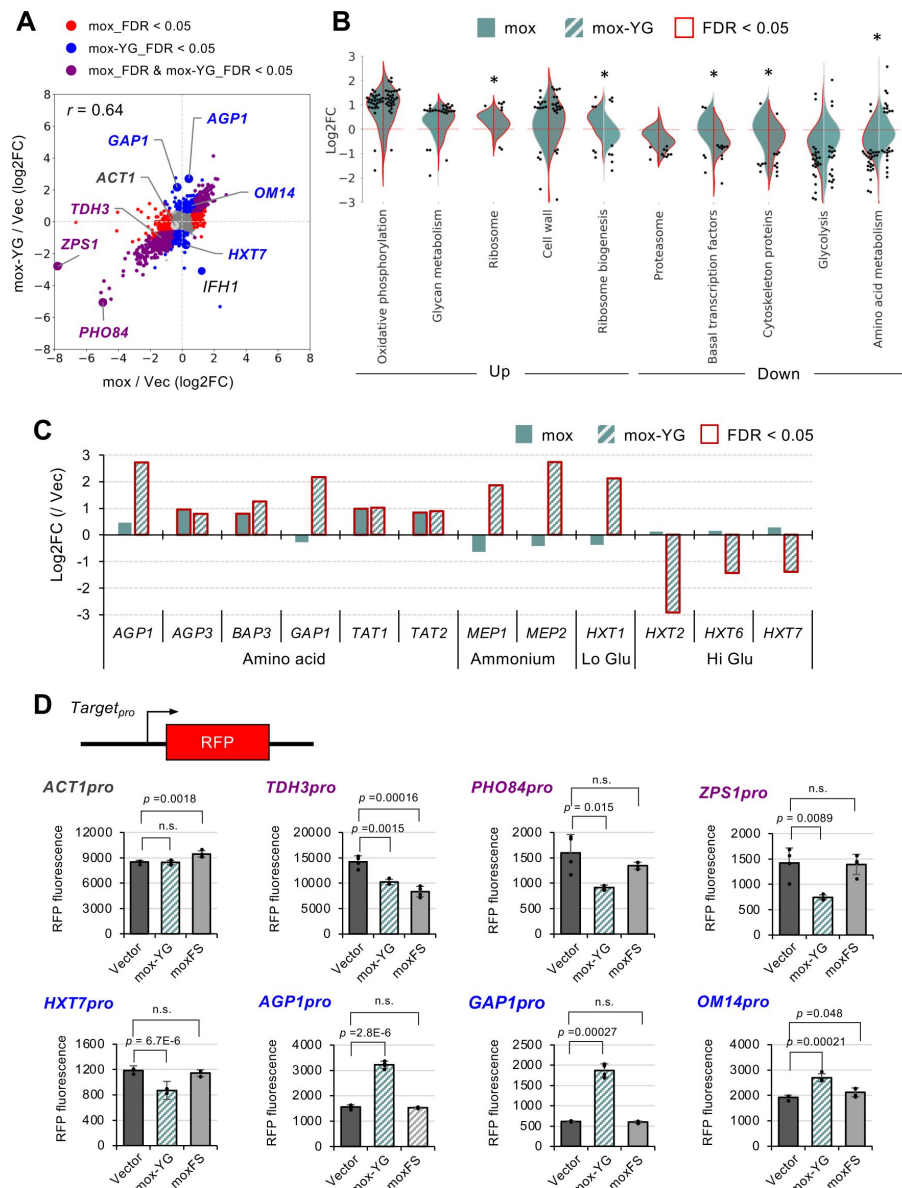


Figure 4.

Transcriptional response of mox and mox-YG overexpression.

A) Comparison of transcriptional responses of mox and mox-YG overexpression. Genes that showed common or different transcriptional responses in mox and mox-YG overexpression are shown in indicated colors. Genes that showed characteristic responses are also shown. FDR: false discovery rate. r : Pearson correlation coefficient.

B) Gene groups that showed significant transcriptional changes. Among the KEGG orthology level 3 categories, gene groups in the category that were significantly up-regulated or down-regulated (FDR < 0.05) in mox or mox-YG overexpression are shown in the violin plots. Genes with significant expression changes (FDR < 0.05) within the same category are shown by swarm plots. Asterisks indicate significant differences (FDR < 0.05) between mox and mox-YG comparisons in gene groups belonging to the same category. Comparisons for all categories are shown in [Figure 4 S1C](#).

C) Expression changes in representative amino acid, ammonium, and glucose transporters.

D) Verification of promoter activity by the reporter assay. Constructs used for promoter analysis with transcription reporters and quantitative results of RFP fluorescence values (arbitrary unit) for each promoter. Bars and error bars indicate the mean and SD of maximum fluorescence values for RFP calculated from four biological replicates. Raw data are shown as dot plots. The p -values were calculated by performing Welch's t -test and applying Bonferroni correction. "n.s." means $p > 0.05$. Detailed constructs and time series data for promoters other than those shown here are in [Figure 4_S2](#).

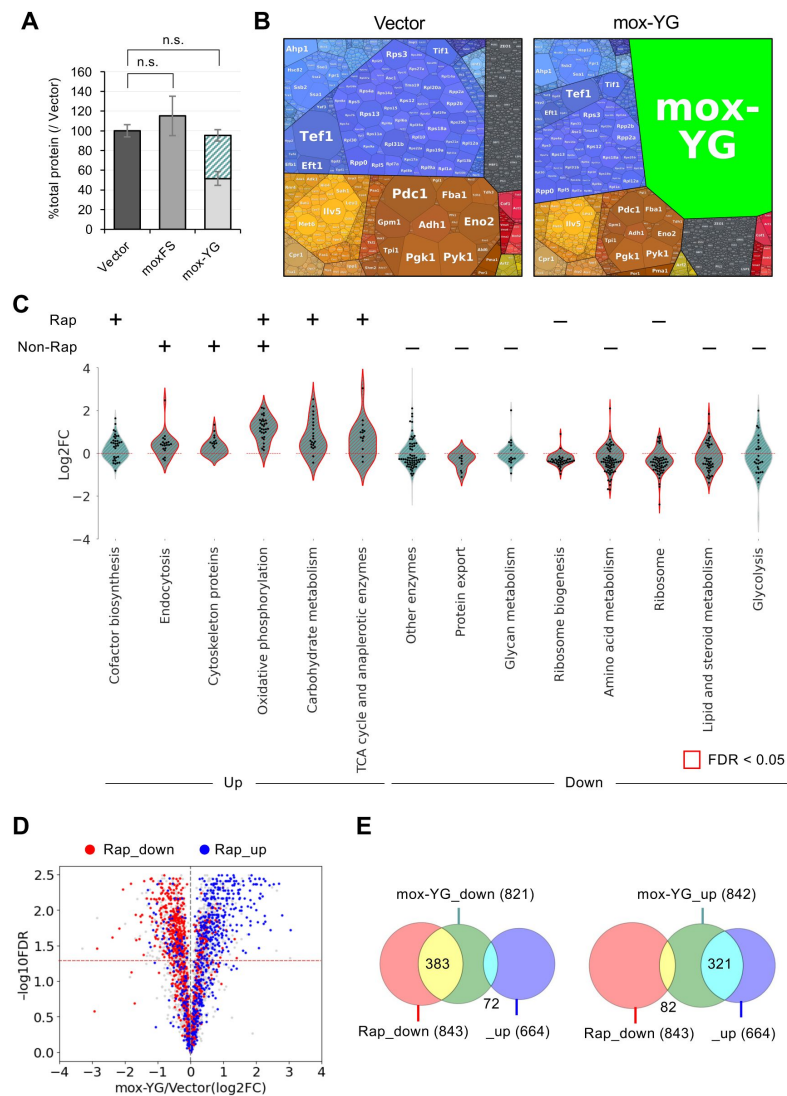


Figure 5.

Overexpression of mox-YG causes a proteome response like that of TORC1 inactivation.

A) Ratio of the total protein amount over the vector. The colored area indicates the amount of overexpressed protein. The amount was calculated from the intensity of the target bands separated by SDS-PAGE. Bars and error bars indicate the mean and SD of protein levels calculated from three biological replicates. The p -values were calculated by performing Welch's t -test and applying Bonferroni correction. "n.s." means $p > 0.05$.

B) Visualization of the proteome of the vector control and mox-YG overexpressing cells using Proteomaps (Liebermeister et al., 2014). Each polygon represents the mass fraction of the protein within the proteome. Similar colors indicate similar or closely related pathways and proteins.

C) Violin plots showing groups of genes in the KEGG orthology level 3 category whose expression was increased or decreased by mox-YG overexpression. Gene groups with significantly altered expression (FDR < 0.05) in mox-YG overexpression are shown in red. Genes with significant expression changes (FDR < 0.05) within the same category are shown by swarm plots. "+" and "-" indicate the categories with significantly increased or decreased expression, respectively, when separated into Rapamycin-responsive (Rap) and Rapamycin-non-responsive (Non-Rap) gene populations. Comparisons for all categories are shown in [Figures 5](#), [S3](#) and [S4](#).

D) Volcano plots showing genes whose expression levels were altered by rapamycin treatment in the proteome results of mox-YG overexpressing cells.

E) Venn diagram representing the overlap of the proteome during mox-YG overexpression and rapamycin treatment. The number of genes in each area is shown.

For C, D, and E, published data (Gowans et al., 2018) was used for comparison with the rapamycin response genes.

responsive (Rap) and rapamycin-non-responsive (Non-Rap) gene populations. Carbohydrate metabolism, elevated TCA cycle, and decreased ribosome synthesis were suggested to be responses due to TORC1 inactivation, while endocytosis, cytoskeleton, protein transport and lipid synthesis, and glycolysis were suggested to be due to responses other than TORC1 inactivation (**Figure 5** [S4](#)). Oxidative phosphorylation included both Rap responses (12/32) and non-responses (20/32) (**Figure 5** [S4C](#)). We confirmed an increase in Rip1, one of the proteins in the oxidative phosphorylation category, using fluorescence microscopy observations (**Figure 5** [S5](#)). Thus, the massive expression of *mox-YG* resulted in a decrease in intracellular proteins and changes in proteome composition. The changes included decreased ribosomes and an increase in mitochondrial proteins the primary cause of which seemed to be the inactivation of the TORC1 pathway due to nitrogen source depletion.

Overexpression of *mox-YG* causes a dysplastic nucleolus, which is alleviated by mutations in the nuclear exosome

Ribosomes are a crucial factor for protein burden because they are themselves protein synthesizers while being synthesized in the highest amounts in the cell. Therefore, we further focused on the behavior of ribosomes in *mox-YG* cells and expression changes were investigated in detail for each ribosomal protein (**Figure 6** [S1](#)). While the majority of the 84 ribosomal proteins detected (45 of them significantly) decreased, six (Rps12, Rps31, Ppp0, Rpp2a, Rpl5, and Rpl12) showed significant increases. Interestingly, all decreased ribosomal proteins are assembled into ribosomes in the nucleolus, while all increased ribosomal proteins are assembled into ribosomes in the cytoplasm (Lafontaine, 2015 [S1](#)). They constitute the “beak” of the 40S subunit and the “P-stalk” and the “central protuberance (pc)” of the 60S subunit (**Figure 6A** [S1](#)). Electron micrographs showed an abnormal nucleolar formation of *mox-YG* overexpressing cells (**Figures 6B** [S1](#), **6_S3**), in addition to reduced ribosome density in the cytoplasm (**Figure 6** [S2A](#)). This abnormal nucleolar formation was also confirmed by fluorescence microscopy images of Nsr1 (Nucleolin), the major protein of the nucleolus (**Figure 6** [S2B-D](#)).

Based on these results, we hypothesized that overexpression of *mox-YG* creates an imbalance between the amount of ribosomal protein synthesized in the cytoplasm and the amount of rRNA synthesized in the nucleolus (**Figure 6C** [S1](#), **D**). The imbalance may lead to the creation of misassembled ribosomes, which are eventually degraded. We speculated that the primary enzyme degrading the misassembled ribosome should be the nuclear exosome. This is because our previous genome-wide mutation analysis identified exosome mutants (*dis3-1*, *mtr3-ts*, *mtr4-1*, *rpn42-ph*, *rpn45-ph*, and *ski6-ph*) as primary mitigators of protein burden (Kintaka et al., 2020 [S1](#)). We confirmed that the *mtr4-1* mutant does not show growth retardation upon *mox-YG* overexpression (**Figure 6E** [S1](#)). In fact, in *mtr4-1* cells, there was no difference in growth rate between the vector and *mox-YG* overexpression (**Figure 6** [S5B](#)), but rather a shorter lag time for the *mox-YG* overexpression than the vector control (**Figure 6E** [S1](#)). We next quantified protein upon *mox-YG* overexpression in *mtr4-1* cells (**Figure 6F** [S1](#)). Compared to the wild-type (Vector), the total protein content in *mtr4-1* (Vector) cells was reduced to 78%, which was likely related to the decreased growth. In the *mtr4-1* *mox-YG* overexpressing cells, the amount of other proteins decreased (to 54%), overwhelmed by the 40% expression of *mox-YG*. However, the overall protein content increased compared to the *mtr4-1* vector control, with a composition like the wild-type *mox-YG* overexpressing cells. Therefore, in *mtr4-1* cells, even though *mox-YG* overexpression reduced the amount of other proteins (to 54%) compared to the vector (78%), the growth rate was not reduced (**Figure 6** [S5B](#)). This suggests that *mox-YG* overexpression restores the effect of growth inhibition caused by *mtr4-1*. Next, we observed the composition of the nucleolus in *mtr4-1* (**Figures 6G** [S1](#), **6_S6B, E**). The failure of nucleolus formation observed upon *mox-YG* overexpression was partially rescued in the *mtr4-1* strain. This supports our hypothesis that misassembled ribosomes are degraded by the exosome (**Figure 6D** [S1](#)), and with low exosome activity, nucleolus formation (and efficient ribosome assembly) would be possible even with overexpression of *mox-YG* (**Figure 6** [S4D](#)).

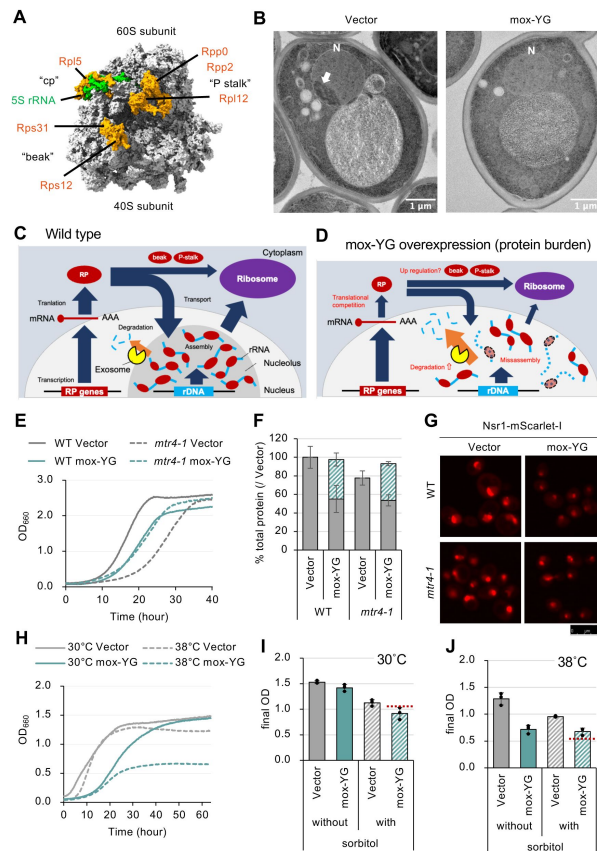


Figure 6.

mox-YG overexpression causes abnormal nucleolus formation.

A. 3D structure of the ribosome. PDB model 6GQV (Pellegrino et al., 2018) is used for the base model and colored using ChimeraX software (Meng et al., 2023). Ribosomal proteins with increased expression upon mox-YG overexpression are shown with the named structures containing them. See also Figure 6_S1 for the quantitative data.

B) Electron microscope images of the vector control and mox-YG overexpressing cells. The arrow in the image points to the nucleolus structure. N: nucleus. Images of other observed cells are shown in **Figure S6_S3**.

C, D) Model diagrams showing the hypothetical situation of the wild type (**C**) and mox-YG overexpressing cells (**D**). In WT cells, enough ribosomal proteins (RPs) are produced, resulting in RP-assembled rRNAs and the formation of nucleolus with normal morphology. On the other hand, in mox-YG overexpressing cells, the amount of RP is reduced due to translation competition, increasing misassembled rRNA. As a result, degradation of rRNA by exosomes may be accelerated, resulting in abnormal nucleolus morphology.

E) Growth curves of WT and *mtr4-1* mutant cells with the control vector or upon overexpression of mox-YG at 30°C. The solid or dotted lines show the average calculated from three biological replicates. Growth curves with SDs of replicates are shown in **Figure 6_S5**.

F) Ratio of total protein levels of WT and *mtr4-1* strains with the vector or upon mox-YG overexpression, calculated the total protein level of WT cells with the control vector as 100%. Gray bars indicate expression levels of proteins other than mox-YG; green shaded bars indicate mox-YG expression levels. Error bars were SDs calculated from three biological replicates.

G) Fluorescence microscopy image of nucleolus-localized protein Nsr1-mScarlet-I of the WT and *mtr4-1* mutant cells with the vector or mox-YG overexpression. Bright field and merged images, and quantification of the nucleolus size are shown in **Figures 6_S5D** and **S5E**.

H) Growth curves of the cells with the vector or under mox-YG overexpression cultivated at 30°C or 38°C. The solid or dotted lines show the average calculated from three biological replicates. Growth curves with SDs of replicates are shown in **Figure 6_S6A**.

I, J) Final OD of the cell culture with or without 1M sorbitol at 30°C (**I**) or 38°C (**J**). The bars and error bars show the means and SDs calculated from three biological replicates. The red dotted line indicates the final OD estimated from the product of (mox-YG without sorbitol) / (Vector without sorbitol) and (Vector with sorbitol) / (Vector without sorbitol). Growth curves with SDs of replicates are shown in **Figures 6_S6C, D**.

While experimenting with mox-YG overexpressing cells, we found that these cells stop growth earlier (lower final density) at higher temperatures (38°C) than the vectors (**Figure 6** [↗](#), **S6A**). Cells at this time did not show cell death such as cell bursting (**Figure 6** [↗](#), **S6**). Recently, it was reported that intracellular osmotic pressure increases at high temperatures, resulting in less formation of liquid-liquid phase-separated structures such as nucleoli, which are recovered when cells are placed in hyperosmotic conditions (Watson et al., 2023 [↗](#)). We hypothesized that the growth inhibition of mox-YG overexpressing cells at high temperatures may be due to the enhanced failure of nucleolus formation, and thus cultured mox-YG overexpressing cells under hyperosmotic pressure. At 30°C, hyperosmolarity caused the same decrease in growth for the vector control and mox-YG overexpression (**Figures 6I** [↗](#), **6_S6C**). On the other hand, at 38°C, hyperosmolarity caused a decrease in growth only in the vector control and no further decrease in mox-YG overexpression cells (**Figures 6J** [↗](#), **6_S6D**). The same trend was observed regarding the size of the nucleosome (**Figures 6** [↗](#), **S6E, F**). Under the conditions with added sorbitol and under mox-YG overexpression, the size of the nucleoli was smaller than the control at 30°C, whereas, at 38°C, it was rather larger than the control. This suggests that mox-YG overexpression and hyperosmolarity may mitigate each other's growth-inhibitory effects, and one explanation may be that hyperosmolarity suppresses the abnormal organization of the nucleolus caused by mox-YG overexpression.

Discussion

In this study, we found that the fluorescent properties of FPs constrain their expression limits. This fluorescence-induced cytotoxicity appears as different phenotypes depending on the FPs; an excess of fluorescent mCherry causes cell enlargement, but such a phenotype was not seen with green fluorescent proteins. On the other hand, the cause of the cytotoxicity is not clear currently. Fluorescence itself does not seem to constrain the expression limit, because the mox expression limit remained the same when cultured in the dark or under strong light (**Figure 2** [↗](#), **S6**). We also believe that H₂O₂ generated during fluorophore maturation (Craggs, 2009 [↗](#)) is unlikely. In the presence of H₂O₂, mox overexpressing cells grew slightly slower than mox-YG overexpressing cells (**Figure 2** [↗](#), **S6B**). However, no transcriptional response to oxidative stress, including the catalase gene (*CTT1*), was seen with mox overexpression (FDR <0.187, over the vector, **Data S1**). Because *CTT1* expression was elevated in mox-YG overexpression (FDR <0.006, over the vector, **Data S1**), presumably due to enhanced respiration, mox-YG overexpression might rather increase oxidative stress tolerance. Conversely, “inactive” proteins might accumulate more without interfering with intracellular functions because both inactive fluorescent proteins and inactive glycolytic enzymes showed higher neutrality (**Figures 2** [↗](#) and **3** [↗](#)). However, our current data did not clarify the mechanism, and investigating more proteins and their variants would be required.

The currently least cytotoxic proteins, non-fluorescent mox-YG and inactive Gpm1 express more than 40% of the total protein, whereas yeast cells can maintain growth under these conditions (**Figures 2** [↗](#) and **3** [↗](#)). This is higher than the prediction in *E. coli*, which stops growing when 37% of the excess protein is expressed (Bruggeman et al., 2020 [↗](#)). This may be due to the very low cytotoxicity of the proteins expressed in this study, or it may be because eukaryotic cells have greater capacity than prokaryotic cells. In the mox-YG overexpressing cells, we observed a previously expected but unobserved cellular physiology of the protein burden: depletion of protein synthesis resources (i.e., nitrogen source; **Figure 4** [↗](#)) and a decrease in other proteins (glycolytic enzymes and ribosomes; **Figure 5** [↗](#)), possibly due to competition for synthesis. In addition, metabolic conversion to mitochondria and failure of nucleolus formation were observed (**Figures 5** [↗](#) and **6** [↗](#)). As a matter of fact, the nitrogen source depletion response (**Figure 4** [↗](#)) itself is not self-evident. Because the total protein content in the mox-YG overexpressing cells is the same as that in the vector control (**Figure 5A** [↗](#)), the intracellular nitrogen source used for protein synthesis should not be depleted. Then, why does the nitrogen source depletion response occur? Excess mox-YG may promote wasteful consumption of amino acids and nitrogen, or there may be

pathways that sense the amount of “useful proteins” and their ability to be synthesized, or there may be depletion of specific amino acids in mox-YG excess. Another possibility is that there may be some mechanism that maintains the amount of total protein in the cell even when the nitrogen source is depleted. Depletion of nitrogen sources probably results in the inactivation of the TORC1 pathway (**Figure 5C-E**), metabolic conversion from fermentation to respiration, and ribosome depression to an economic mode. In *E. coli*, the ppGpp pathway plays this role (Shachrai et al., 2010). While the above is thought to be a physiological response to protein burden, a defect in ribosome synthesis in the nucleolus was also suggested (**Figure 6B**). This could be due to a conflict between nuclear rRNA synthesis and cytoplasmic protein synthesis or to the negative effects of the inability to create a liquid-liquid phase-separated structure due to dilution effects (see below). The partial restoration of nucleolus formation by a mutation of the exosome (**Figure 6G**) supports this possibility.

When one type of protein is expressed as much as 40% of the total, as in the present study, the cytoplasmic crowding, viscosity, pH, etc. may change due to the nature of the protein and the effect of the reduced protein by competition. These may alter the movement of substances. It is known that when ribosomes are reduced, the fluidity of large proteins changes (Delarue et al., 2018). Water activity-osmotic pressure may change (Watson et al., 2023), and pH may also be affected. These may affect the formation of liquid-liquid phase-separated structures (Delarue et al., 2018; Watson et al., 2023). The failure of nucleolus formation observed in mox-YG overexpression (**Figure 6B**) may be due to such a condition change. In fact, the growth of mox-YG overexpressing cells is significantly impaired at high temperatures (**Figure 6H**), where liquid-liquid phase separation is less likely to occur (Watson et al., 2023), while the effect appears to be reduced at high osmolarity (**Figure 6J**), where it is more likely to occur (Watson et al., 2023). A mutation of the nuclear exosome restored nucleolus formation (**Figure 6G**), which might be due to the reformation of the liquid-liquid phase-separated structure (i.e., nucleolus) by increasing the concentration of rRNA. These have not been considered phenomena that occur with protein overexpression. The engineering of physicochemical properties of the cytoplasm thus may be possible by mass expression of proteins with specific properties.

In this study, in addition to mox-YG, inactive Gpm1 was also expressed in large amounts as well (**Figure 3**). Both are globular proteins with relatively small molecular weight, but it is not yet clear what characteristics are necessary for them not to be cytotoxic. Further investigation of the neutrality of various proteins will clarify these features. In this study, expression limits were investigated, but no experiments were performed to gradually increase expression levels. For example, it would be meaningful to investigate at what expression level of mox-YG a shift to the economy mode or failure of nucleolus formation occurs. Also, due to the limitations of the experimental system, it may not be possible to express mox-YG to its true limit. The expression of one specific protein cannot be 100% of the total protein, and there is always a limit. As the number of ribosomes continues to decrease due to competition for synthesis, the excess proteins cannot be synthesized either. Thus, there must be a final equilibrium point. A more powerful and controllable expression system would be needed to investigate this. Although the translation system appears to be the primary target of burden in mox-YG overexpression (**Figure 2**, **S5C, D**), the bottleneck in the expression system may also differ depending on gene design (Hausser et al., 2019).

Materials and Methods

Strains, growth conditions, and yeast transformation

BY4741 (*MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0*) (Brachmann et al., 1998) was used as the host strain for the experiments. Yeast culture and transformation were performed as previously described (Amberg et al., 2005). A synthetic complete (SC) medium without uracil (Ura) or

leucine (Leu), as indicated, was used for yeast culture. The strains used in the reporter assay are listed in the Key Resource Table.

Plasmid construction

The plasmids were constructed by homologous recombination activity of yeast cells (Oldenburg 1997), and their sequences were verified by DNA sequencing. Plasmids used in this study are listed in the Key Resource Table.

Genetic tug-of-war

To overexpress a target protein to a level that causes growth inhibition (in this study this is referred to as “expression limit”), we used gTOW (**Figure 2** [S1A](#)) (Moriya et al., 2012 [S1A](#), 2006 [S1A](#)). The gene of the target protein is incorporated into the gTOW plasmid (here, pTOW40836) and introduced into yeast cells lacking *ura3* and *leu2* genes. The selection marker in this case is uracil (–Ura conditions). Since this plasmid has the replication origin of 2 μ m plasmid (2 μ *ORI*), this plasmid becomes multicopy in the cell (about 30 in the case of the vector). When yeast cells carrying the plasmid are transferred to a medium without leucine and uracil (–Leu/Ura conditions), the copy number of the plasmid in the cells increases (to about 120 in the case of the vector). This is because *leu2-89* on the plasmid is a *LEU2* allele with a large deletion in its promoter, so cells with higher plasmid copy numbers will grow faster in –Leu/Ura conditions. In other words, *leu2-89* acts as a bias to raise the plasmid copy number in –Leu/Ura conditions. If overexpression of the target protein inhibits growth, the plasmid copy number will be lower than the copy number that gives rise to the critical expression of that protein. In other words, the target gene acts as a bias to lower the plasmid copy number. The resulting tug-of-war between the selection biases created by the two genes leads to a target protein being expressed at close to the critical expression level in –Leu/Ura conditions (the green-colored area in **Figure 2** [S1A](#)).

Measuring growth rate

Cellular growth was measured by monitoring OD₆₆₀, every 10 min using a compact rocking incubator (TVS062CA, ADVANTEC). The max growth rate (MGR) was calculated as described previously (Moriya et al. 2006 [S1A](#)). Average values, SD, and *P*-values of Welch’s *t*-test were calculated from at least three biological replicates. The growth data of **Figures 2** [S9A](#), **2** [S9B](#), and **4** [S2](#) was measured by monitoring OD₅₉₅, respectively, every 10 min using a microplate reader (Infinite F200, TECAN).

Protein analysis

BY4741 cells overexpressing a target protein were cultivated in SC–Leu/Ura medium. The total protein was extracted from log-phase cells (OD₆₆₀ = 1.0) with a NuPAGE LDS sample buffer (Thermo Fisher Scientific) after 0.2 mol/l NaOH treatment (Kushnirov 2000). For each analysis, the total protein extracted from 1 optical density (OD) unit of cells with OD₆₆₀ was used. For total protein visualization, the extracted total protein was labeled with Ezlabel FluoroNeo (ATTO), as described in the manufacturer’s protocol, and separated by 4–12% sodium-dodecyl sulfate acrylamide gel electrophoresis (SDS-PAGE). Proteins were detected and measured using the LAS-4000 image analyzer (GE Healthcare) in SYBR–green fluorescence detection mode and Image Quant TL software (GE Healthcare). Protein quantification was performed as shown in **Figure 2** [S1B](#). Average values, SD, and *P*-values of Welch’s *t*-test were calculated from three biological replicates.

Measuring GFP fluorescence

GFP fluorescence (Ex485nm/Em535nm) was measured every 10 min using a microplate reader (Infinite F200, TECAN). Average values, SD, and *p*-values of Welch’s *t*-test were calculated from eight biological replicates.

RNA-seq analysis

BY4741 cells overexpressing a target protein were cultured in SC–Leu/Ura medium and harvested at the logarithmic growth phase. RNA extraction was performed according to (Köhler and Domdey, 1991). The purified RNA was quality checked by BioAnalyzer (Agilent) or MultiNA (Shimadzu), and the concentration was measured by Qubit (Thermo Fisher Scientific). Purified RNA was stored at –80°C until subsequent experiments. cDNA libraries were prepared using the TruSeq Stranded Total RNA kit (Illumina), using half the protocol of the TruSeq RNA library prep kit. Three biological replicates were analyzed for all strains. The sequences were checked for read quality by FastP (Chen et al., 2018) and then aligned using Hisat2 (Kim et al., 2019). The aligned data were formatted into bam files by Samtools (Li et al., 2009) and quantified by StringTie (Pertea et al., 2015). Finally, expression level variation analysis was performed by EdgeR (Robinson et al., 2010). The raw data were deposited into DDBJ (accession number: PRJDB18064). The processed data (**Data_S1_RNAseq.csv**) included the average expression levels (log10CPM) of 6685 genes in the vector control, the changes in expression (log2FC) upon overexpression of *mox* or *mox-YG* compared to the vector control, and the significance values of expression change (*mox_FDR*, *mox-YG_FDR*). **Figures 4_S1A, B** is the visualization of this data using volcano plots. This data is used in subsequent data analyses below.

Reporter assay

The strains used for the reporter assay were created by introducing the promoter region of each gene and the sequence of mScarlet-I, a type of RFP, into the *FCY1* locus of the genome as shown in **Figure 4_S2A**. Correct integration was checked by histidine prototrophy and 5-FC resistance of the strain. The cell density (OD₅₉₅) and RFP fluorescence (Ex535nm/Em590nm) of the created strains were measured every 10 min using a microplate reader (Infinite F200PRO, TECAN). Average and SD of max RFP fluorescence, and p-values of Welch's *t*-test were calculated from eight biological replicates.

Proteome analysis

The vector control and *mox-YG* overexpressing cells (three biological replicates) were cultured in 5 ml of SC–LeuUra medium with shaking and cells were collected at OD₆₆₀ = 1. Cells were washed twice with PBS, frozen on dry ice, and transported to Kazusa DNA Research Institute. The samples were prepared as described in the previous study (Hughes et al., 2019; Kawashima et al., 2022). The cells were dissolved in 100 mM Tris-HCl (pH 8.0) containing 4% SDS and 20 mM NaCl using BIORUPTOR BR-II (SONIC BIO Co., Kanagawa, Japan). The 20 µg of extracted proteins were quantified using Pierce™ BCA Protein Assay Kit (Thermo Fisher Scientific, WA, USA) at 500 ng/µL. The protein extracts were reduced with 20 mM tris (2-carboxyethyl) phosphine for 10 min at 80 °C followed by alkylation with 30 mM iodoacetamide for 30 min at room temperature in the dark. Protein purification and digestion were performed using the sample preparation (SP3) method (Kawashima et al., 2022). The tryptic digestion was performed using 500 ng/µL Trypsin/Lys-C Mix (Promega, Madison, WI, USA) for overnight at 37 °C. The digests were purified using GL-Tip SDB (GL Sciences, Tokyo, Japan) according to the manufacturer's protocol. The peptides were dissolved again in 2% ACN containing 0.1% TFA and quantified using BCA assay at 150 ng/µL. The digested peptides were loaded directly using a 75 µm × 12 cm nanoLC nano-capillary column (Nikkoy Technos Co., Ltd., Tokyo, Japan) at 40°C and then separated with a 30-minute gradient (mobile phase A = 0.1% FA in water, B = 0.1% FA in 80% ACN) consisting of 0 min 8% B, 30 min 70% B at a flow rate of 200 nL/min using an UltiMate 3000 RSLCnano LC system (Thermo Fisher Scientific). The eluted peptides were detected using a quadrupole Orbitrap Exploris 480 hybrid mass spectrometer (Thermo Fisher Scientific) with normal window DIA. The MS1 scan range was set as a full scan with *m/z* 490–745 at a mass resolution of 15,000 to set an Auto Gain Control (AGC) target for MS1 as 3×10^6 and a maximum injection time of 23 msec. The MS2 were collected at more than *m/z* 200 at 30,000 resolutions to set an AGC target of 3×10^6 , maximum injection time of

“auto”, and fixed normalized collision energy of 28%. The isolation width for MS2 was set to 4 m/z, and for the 500-740 m/z window pattern, an optimized window arrangement was used in Scaffold DIA (Proteome Software, Inc., Portland, OR, USA). The raw data were searched against an *in silico* predicted spectral library using DIA-NN (Demichev et al., 2020 [DOI](#)). (version:1.8.1, <https://github.com/vdemichev/DiaNN> [DOI](#)) First, an *in-silico* predicted spectral library was generated from the human protein sequence database (UniProt id UP000005640, reviewed, canonical,

20381 entries) using DIA-NN. The DIA-NN search parameters were as follows: protease, trypsin; missed cleavages, 1; peptide length range, 7-45; precursor charge range, 2-4; precursor mass range, 495-745; fragment ion m/z range, 200-1800; mass accuracy, 10 ppm; static modification, cysteine carbamidomethylation; enabled “Heuristic protein interferences,” “Use isotopologues,” “MBR,” and “No shared spectra.” Additional commands were set as follows: “mass acc cal 10,” “peak translation,” and “matrix spec q”. The protein identification threshold was set at < 1% for both peptide and protein false discovery rates (FDRs). The signal intensity data for 4241 proteins (Data_S2_Proteome.csv) was used for further data analysis below.

Data analysis

The analysis and visualization of RNAseq data and proteome data were conducted using custom Python code, and coding and execution were carried out in the Collaboratory. In the RNAseq data, 6,685 protein-coding genes were detected (Data_S1_RNAseq.csv). Among these, 4,291 genes assigned to various categories of KEGG Orthology level 3 (Kanehisa et al., 2023 [DOI](#)) were placed into 77 categories. Analysis was performed on 47 categories that contained more than 10 gene elements. We investigated whether the expression of genes in each category, as a group, was significantly higher or lower upon overexpression of *mox* and *mox*-YG compared to the vector. Specifically, *mox*/vector (log2FC) and *mox*-YG/vector (log2FC) were calculated along with their FDR values, and each gene was assigned to a KEGG orthology level 3 category. Next, whether the gene groups within each category showed significantly higher or lower expression changes compared to gene groups outside of their category was tested using the Mann-Whitney U test. Significant differences were indicated in the graphs with a red border and red stripes. The graphs showed the overall distribution of genes within each category using violin plots, and only genes showing significant differences in *mox*/vector (log2FC) and *mox*-YG/vector (log2FC) were highlighted in swarm plots. An asterisk indicated significant differences on the graphs when the expression changes (log2FC) between *mox*/vector and *mox*-YG/vector were significant by the Mann-Whitney U test. The significance threshold was set at FDR < 0.05 for all. Only categories where either *mox* or *mox*-YG overexpression showed significant expression changes were extracted and presented as **Figure 4B** [DOI](#), and all 47 categories were shown in **Figure 4** [DOI](#)_S1.

The proteome analysis was conducted using the signal intensity data (Data_S2_proteome.csv). This data includes proteins that were detected only in some samples under certain conditions or replicates, especially those with low signals or large expression variations. Additionally, the total protein signal sum excluding *mox*-YG was almost the same between the vector control and *mox*-YG overexpression (**Figure 5** [DOI](#)_S2A). This result contradicts the SDS-PAGE findings, which indicate that the total protein mass per cell, including *mox*-YG, is the same between the vector control and *mox*-YG overexpression (**Figure 5A** [DOI](#)). Therefore, it is considered necessary to adjust the total protein signal sum, including *mox*-YG, to match that of the vector control to understand the proteome change per cell. In this study, however, we used the most conservative approach by removing proteins with zero values and not applying any overall corrections (**Figure 5** [DOI](#)_S2B), which allowed the analysis of 3,588 proteins. Although this may not correctly analyze “per cell” proteome change, it enables us to capture the change in proteome composition (excluding *mox*-YG), thus providing information about cellular responses. If the total correction is applied, only 138 proteins were increased (**Figure 5** [DOI](#)_S2B), most of which are mitochondrial proteins.

For the detected 3,588 proteins above, we compared the increase or decrease of protein groups within each category of KEGG Orthology Level 3 with those outside the category. Analysis was performed on 45 categories, each containing more than 10 proteins, to investigate whether the expression of protein groups in each category was significantly higher or lower under mox-YG overexpression compared to the vector control. Specifically, the average log2FC and its FDR values were calculated using data from three replicates for mox-YG/vector, and each protein was assigned to a KEGG orthology level 3 category. Next, whether the protein groups within each category showed significantly higher or lower expression changes compared to protein groups outside of their category was tested using the Mann-Whitney U test. Significant differences were indicated in the graphs with a red border and red stripes. The graphs showed the overall distribution of proteins within each category using violin plots, and only proteins showing significant differences in mox-YG/vector (log2FC) were highlighted in swarm plots. The significance threshold was set at FDR < 0.05 for all. Only categories where mox-YG overexpression showed significant expression changes were extracted and presented in **Figure 5C** and all 47 categories were shown in **Figure 5_S3**. Published data (Gowans et al., 2018) was used to separate protein groups into rapamycin-responsive and non-responsive, and similar analysis and visualization were conducted (**Figures 5_S4A, B**).

Fluorescence microscopic observation

Cells overexpressing target proteins were cultured in SC-LeuUra medium and observed under an inverted microscope (DMI 6000 B, Leica) at log phase (OD₆₆₀ = 1.0). Images were acquired with Leica Application Suite and processed with Leica Application Suite X software (Leica Microsystems). Cells in brightfield images were separated from the images using yeast_segmentation-master_v3 (Lu et al., 2019). Cell morphological traits were assessed using parameters obtained when analysed using CellProfiler (Ver. 4.2.6) (Carpenter et al., 2006). “AreaShape_Area” was used to analyse FP-overexpressing cell size in **Figure 2_S5B**, Nsr1-GFP in **Figure 6_S2C** and Nsr1-mScarlet-I in **Figure 6_S5E, S6F**. For analysis of the fluorescence intensity of Rip1-mScarlet-I in **Figure 5_S5B**, “Intensity_IntegratedIntensity” was used. Intensity_MeanIntensity” was used to analyze the fluorescence intensity of Nsr1-GFP in **Figure 6_S2D**. Ten images per experiment were used for analysis.

Electron microscopic observation

Cells grown to log phase (OD₆₀₀ = 1.0) at 30°C in SC-LeuUra medium were collected and transported to Tokai Electron Microscope Co. Transported samples were sandwiched between copper plates, flash-frozen with liquefied propane, and then dehydrated with anhydrous ethanol. cells were fixed with a 5:5 mixture of propylene oxide and resin (Quetol-651, Nissin EM Co.). Fixed cells were cut with an ultramicrotome (Ultracut UCT, Leica) to prepare 80 nm thick sections, stained with 2% uranyl acetate and lead stain solution (Sigma-Aldrich Co.), and observed with a transmission electron microscope (JEM-1400 Plus, JEOL Ltd.).

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Reviewer #1 (Public Review):

Summary:

The study "Impact of Maximal Overexpression of a Non-toxic Protein on Yeast Cell Physiology" by Fujita *et al.* aims to elucidate the physiological impacts of overexpressing non-toxic proteins in yeast cells. By identifying model proteins with minimal cytotoxicity, the authors claim to provide insights into cellular stress responses and metabolic shifts induced by protein overexpression.

Strengths:

The study introduces a neutrality index to quantify cytotoxicity and investigates the effects of protein burden on yeast cell physiology. The study identifies mox-YG (a non-fluorescent fluorescent protein) and Gpm1-CCmut (an inactive glycolytic enzyme) as proteins with the lowest cytotoxicity, capable of being overexpressed to more than 40% of total cellular protein while maintaining yeast growth. Overexpression of mox-YG leads to a state resembling nitrogen starvation probably due to TORC1 inactivation, increased mitochondrial function,

and decreased ribosomal abundance, indicating a metabolic shift towards more energy-efficient respiration and defects in nucleolar formation.

Weaknesses:

While the introduction of the neutrality index seems useful to differentiate between cytotoxicity and protein burden, the biological relevance of the effects of overexpression of the model proteins is unclear.

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Reviewer #2 (Public Review):

Summary:

In this manuscript, Fujita et al. characterized the neutrality indexes of several protein mutants in *S. cerevisiae* and uncovered that *mox-YG* and *Gpm1-CCmut* can be expressed as abundant as 40% of total proteins without causing severe growth defects. The authors then looked at the transcriptome and proteome of cells expressing excess *mox-YG* to investigate how protein burden affects yeast cells. Based on RNA-seq and mass-spectrometry results, the authors uncover that cells with excess *mox-YG* exhibit nitrogen starvation, respiration increase, inactivated TORC1 response, and decreased ribosomal abundance. The authors further showed that the decreased ribosomal amount is likely due to nucleoli defects, which can be partially rescued by nuclear exosome mutations.

Strengths:

Overall, this is a well-written manuscript that provides many valuable resources for the field, including the neutrality analysis on various fluorescent proteins and glycolytic enzymes, as well as the RNA-seq and proteomics results of cells overexpressing *mox-YG*. Their model on how *mox-YG* overexpression impairs the nucleolus and thus leads to ribosomal abundance decline will also raise many interesting questions for the field.

Weaknesses:

The authors concluded from their RNA-seq and proteomics results that cells with excess *mox-YG* expression showed increased respiration and TORC1 inactivation. I think it will be more convincing if the authors can show some characterization of mitochondrial respiration/membrane potential and the TOR responses to further verify their -omic results.

In addition, the authors only investigated how overexpression of *mox-YG* affects cells. It would be interesting to see whether overexpressing other non-toxic proteins causes similar effects, or if there are protein-specific effects. It would be good if the authors could at least discuss this point considering the workload of doing another RNA-seq or mass-spectrum analysis might be too heavy.

<https://doi.org/10.7554/eLife.99572.1.sa1>

Reviewer #3 (Public Review):

Summary:

Protein overexpression is widely used in experimental systems to study the function of the protein, assess its (beneficial or detrimental) effects in disease models, or challenge cellular systems involved in synthesis, folding, transport, or degradation of proteins in general. Especially at very high expression levels, protein-specific effects and general effects of a high

protein load can be hard to distinguish. To overcome this issue, Fujita et al. use the previously established genetic tug-of-war system to identify proteins that can be expressed at extremely high levels in yeast cells with minimal protein-specific cytotoxicity (high 'neutrality'). They focus on two versions of the protein *mox*-GFP, the fluorescent version and a point mutation that is non-fluorescent (*mox*-YG) and is the most 'neutral' protein on their screen. They find that massive protein expression (up to 40% of the total proteome) results in a nitrogen starvation phenotype, likely inactivation of the TORC1 pathway, and defects in ribosome biogenesis in the nucleolus.

Strengths:

This work uses an elegant approach and succeeds in identifying proteins that can be expressed at surprisingly high levels with little cytotoxicity. Many of the changes they see have been observed before under protein burden conditions, but some are new and interesting. This work solidifies previous hypotheses about the general effects of protein overexpression and provides a set of interesting observations about the toxicity of fluorescent proteins (that is alleviated by mutations that render them non-fluorescent) and metabolic enzymes (that are less toxic when mutated into inactive versions).

Weaknesses:

The data are generally convincing, however in order to back up the major claim of this work - that the observed changes are due to general protein burden and not to the specific protein or condition - a broader analysis of different conditions would be highly beneficial.

Major points:

- (1) The authors identify several proteins with high neutrality scores but only analyze the effects of *mox*/*mox*-YG overexpression in depth. Hence, it remains unclear which molecular phenotypes they observe are general effects of protein burden or more specific effects of these specific proteins. To address this point, a proteome (and/or transcriptome) of at least a *Gpm1*-CCmut expressing strain should be obtained and compared to the *mox*-YG proteome. Ideally, this analysis should be done simultaneously on all strains to achieve a good comparability of samples, e.g. using TMT multiplexing (for a proteome) or multiplexed sequencing (for a transcriptome). If feasible, the more strains that can be included in this comparison, the more powerful this analysis will be and can be prioritized over depth of sequencing/proteome coverage.
- (2) The genetic tug-of-war system is elegant but comes at the cost of requiring specific media conditions (synthetic minimal media lacking uracil and leucine), which could be a potential confound, given that metabolic rewiring, and especially nitrogen starvation are among the observed phenotypes. I wonder if some of the changes might be specific to these conditions. The authors should corroborate their findings under different conditions. Ideally, this would be done using an orthogonal expression system that does not rely on auxotrophy (e.g. using antibiotic resistance instead) and can be used in rich, complex mediums like YPD. Minimally, using different conditions (media with excess or more limited nitrogen source, amino acids, different carbon source, etc.) would be useful to test the robustness of the findings towards changes in media composition.
- (3) The authors suggest that the TORC1 pathway is involved in regulating some of the changes they observed. This is likely true, but it would be great if the hypothesis could be directly tested using an established TORC1 assay.
- (4) The finding that the nucleolus appears to be virtually missing in *mox*-YG-expressing cells (Figure 6B) is surprising and interesting. The authors suggest possible mechanisms to explain this and partially rescue the phenotype by a reduction-of-function mutation in an exosome subunit. I wonder if this is specific to the *mox*-YG protein or a general protein burden effect,

which the experiments suggested in point 1 should address. Additionally, could a mox-YG variant with a nuclear export signal be expressed that stays exclusively in the cytosol to rule out that mox-YG itself interferes with phase separation in the nucleus?

Minor points:

(5) It would be great if the authors could directly compare the changes they observed at the transcriptome and proteome levels. This can help distinguish between changes that are transcriptionally regulated versus more downstream processes (like protein degradation, as proposed for ribosome components).

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